**Mapping (Version 1.0)**

```
Crypto_AEAD_GenerateEncrypt(K, P, A, N) :=
     byte[lengthInBytes(P) + CRYPTO_AEAD_MIC_LENGTH_BYTES] AES-CCM-GenerateEncrypt(K
:= K, P := P, A := A, N := N, Tlen := CRYPTO_AEAD_MIC_LENGTH_BITS)
```

`AES-CCM-GenerateEncrypt()` SHALL be the function described in Section 6.1 of NIST 800-38C with the counter generation function of Appendix A.3 of NIST 800-38C and the formatting function as defined in Appendix A.2 of NIST 800-38C; returns the encoding of the ciphertext and the tag of length `Tlen` bits, as specified in Section 6.1 of NIST 800-38C as a byte array.

### 3.6.2. Decrypt and verify

**Function and description**

```
{boolean success, byte[lengthInBytes(P)] payload}
Crypto_AEAD_DecryptVerify(
     SymmetricKey K,
     byte[lengthInBytes(P) + CRYPTO_AEAD_MIC_LENGTH_BYTES] C,
     byte[] A,
     byte[CRYPTO_AEAD_NONCE_LENGTH_BYTES] N)
```

Performs the decrypt and verify computation on the combined ciphertext and tag `C` and the associated data `A` using the key `K` and a nonce `N`. Note that the encoding of `C` depends on the mapping of the specific instance of the cryptography primitive.

This function has two outcomes:

- If tag verification succeeds, the `success` output is `TRUE` and the `payload` array contains the decrypted payload `P`.
- If tag verification fails, the `success` output is `FALSE` and the contents of the `payload` array is undefined.

 